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Hurricanes Cause Population Expansion of the Branching Coral *Acropora palmata* (*Scleractinia*): Wound Healing and Growth Patterns of Asexual Recruits

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With 8 figures and 1 table

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Abstract. Three mechanisms aiding recovery and expansion of a population of *Acropora palmata* on a patch reef that was directly in the path of Hurricane Andrew were documented: rapid wound healing, high rates of asexual recruitment, and rapid growth rate of the new recruits. In addition, the growth pattern of new recruits was assessed in order to quantify the initial sequence of structural changes during the transition from recruit to adult morphology. Wound healing was initially rapid (1.59 cm of linear growth per month), but slowed with time. Nine months after the storm, 72% of 218 hurricane-generated fragments of *A. palmata* had cemented to the bottom, becoming new asexual recruits. Within 18 months of the storm, the number of fragments in the same area had increased to 271, average fragment size was larger, and 94% had become recruits. Recruits showed complex patterns of branch formation, including rapid growth of solitary proto-branches, differential growth of proto-branches dependent on recruit orientation, and dominance within aggregates of proto-branches. We present a conceptual model that suggests *A. palmata* is adapted to disturbances of both low intensity and high frequency (conditions typical of reef flat zones) and episodic high intensity and low frequency events (hurricanes and tropical storms).

Problem

Disturbance is an important factor structuring coral reefs (GLYNN, 1976; CONNELL, 1978; HAY, 1981; HUGHES & JACKSON, 1985; HUSTON, 1985; DONE, 1992; WITMAN, 1992; BYTHELL *et al.*, 1993). Many types of disturbances to coral reefs have been documented across a wide range of spatial and temporal scales (*e.g.*, KNOWLTON *et al.*, 1981, 1990; PEARSON, 1981; POTTS, 1983; STODDART, 1985; WILLIAMS & BUNKLEY-WILLIAMS, 1990; GLYNN, 1993; ROGERS, 1993). One of the main objectives of recent coral reef research has been to establish links between the frequency and

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intensity of disturbances and the magnitude and timing of the effects in order to understand how disturbance affect different resident populations, community structure, and ultimately overall reef development.

Disturbance is believed not only to be a major source of temporal and spatial heterogeneity in the structure and dynamics of communities, but to be an agent of natural selection in the evolution of natural populations (SOUSA, 1984). Consequently, the disturbance regime experienced by a community may act as a key factor in determining the life-history strategies of its components. ENDEAN & CAMERON (1990) divide corals into two groups based on life-history strategies, opportunists and persisters. Traits associated with opportunists include large numbers of propagules at frequent or timely intervals, high recruitment rates, rapid colony growth, short generation time, high adult mortality, a clumped distribution, and unstable local populations. Characteristics associated with persisters are slower colonization, low recruitment, slow growth, resistance to biological or physical disturbance, and dispersed distribution.

Acropora palmata, a branching, upright coral with a tree-like morphology does not fall readily into this classification scheme as it has characteristics of both opportunists and persisters. Characteristics of *A. palmata* attributed to opportunistic species include fast growth and regeneration rates (GLADFELTER *et al.*, 1978; BAK & CRIENS, 1981; HIGHSMITH, 1982). However, *A. palmata* has limited sexual recruitment capabilities (DUSTAN, 1977; RYLAARSDAM, 1983; ROSESMYTH, 1984) and allocates much energy to structural components (BAK & ENGEL, 1979), characteristics attributed to persisters.

Studies of *A. palmata* distribution and abundance suggest this species may be adapted to chronic levels of intermediate intensity physical disturbance as encountered in the shallow, exposed areas of reefs where it dominates (GOREAU, 1959; GEISTER, 1977; GLADFELTER *et al.*, 1978; ROGERS *et al.*, 1982; JAPP, 1984; DUSTAN, 1985; BYTHELL *et al.*, 1991). *A. palmata* has several adaptations that enable it to persist in high-energy environments, including a strong skeleton, a thick basal attachment to the substratum, and branches usually oriented parallel to the prevailing currents (GRAUS *et al.*, 1977; SCHUHMACHER & PLEWKA, 1981; GLADFELTER, 1991).

In some cases, *A. palmata* has been found to be extremely susceptible to high intensity but low frequency disturbances, such as hurricanes or tropical storms. Significant fragmentation and dislodgment were reported after Hurricanes Edith (GLYNN *et al.*, 1964), Gerta (HIGHSMITH *et al.*, 1980), Allen (WOODLEY *et al.*, 1981), David and Frederic (ROGERS *et al.*, 1982), Hugo (GLADFELTER, 1991), and Andrew (FONG & LIRMAN, 1994). During Hurricane Allen, up to 95% of the existing *A. palmata* cover in transects over the reefs of Discovery Bay, Jamaica, was lost (WOODLEY *et al.*, 1981).

Despite documented damage, there are indications that *A. palmata* is adapted to physical disturbance associated with hurricanes (JACKSON & HUGHES, 1985). The major propagation mode of *A. palmata* in the Caribbean is colony fragmentation (HIGHSMITH, 1982; RYLAARSDAM, 1983; WALLACE, 1985), and it has been hypothesized that local dominance may be achieved by regeneration and subsequent rapid growth rates of hurricane-generated fragments (HIGHSMITH, 1982; HUSTON, 1985; PORTER, 1987), enabling *A. palmata* to re-cover and re-populate intensely disturbed areas (HIGHSMITH *et al.*, 1982; ROGERS *et al.*, 1982; LIRMAN & FONG, 1996a).

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The earliest stages of asexual recruitment of *Acropora palmata* have been documented. The deposition of live fragments of *A. palmata* has been reported after almost every major hurricane in the Caribbean (GLYNN *et al.*, 1964; HIGHSMITH *et al.*, 1980; WOODLEY *et al.*, 1981; ROGERS *et al.*, 1982; GLADFELTER, 1991; FONG & LIRMAN, 1994). Early survivorship rates of hurricane-generated fragments from 35-60% have been measured, with a strong relationship between survivorship and fragment size (HIGHSMITH *et al.*, 1980; ROGERS *et al.* 1982). In addition, BAK & CRIENS (1981) found that stabilization of fragments was dependent on substrate type; there was a significant relationship between substrate type and both fragment survivorship and fusion. However, in only a few studies was the upward growth rate of new branches forming on hurricane-damaged colonies measured (ROGERS *et al.*, 1982; GLADFELTER, 1991).

For *A. palmata* recruits, quickly establishing upward growth leading to the adult morphology is very important as the arboreal shape may confer an advantage in competition for light (SHINN, 1972; PORTER, 1976); without this, *A. palmata* is an inferior direct competitor for space (LANG, 1973; JACKSON, 1991). However, the early structural changes that take place during the transition from a relatively flat fragment to the complex tree-like structure typical of an adult *A. palmata* are largely unknown.

The objective of this study was to determine initial recovery rates and processes for the storm-damaged population of *A. palmata* on Elkhorn reef. Three approaches were used. First, we measured the pattern and rate of wound-healing on colonies that were still standing after the storm. Second, temporal changes in the size-frequency distributions of fragments and new recruits were measured in order to quantify rates of sexual recruitment. Last, the pattern of initial growth of new recruits was assessed in order to quantify the initial sequence of structural changes during the transition from recruit (flat) to adult (tree-like) morphology. This was accomplished by sequential measures of the abundance, height, and spatial distribution of newly forming branches. We tested the hypotheses that the orientation and surface area of new recruits would significantly affect rates of early growth and survivorship.

Material and Methods

1. Study site

Elkhorn reef (25° 21.766′N, 80° 09.961′W), in the northern region of the Florida Reef Tract (Fig. 1), has been characterized as a transitional or embryonic bank reef (JAAP, 1984). It measures approximately 4.5 hectares and is surrounded by sand plains and seagrass beds. Although it is a mid-shore formation located 2.5 km from the outer reef, it is afforded little protection from wave action as the fore-reef is discontinuous in the direction of the prevailing wind (east and south-east; SHINN *et al.*, 1989). There is no typical reef crest formation, but a relatively wide and shallow reef flat.

On 24 August 1994, Hurricane Andrew, a compact and quickly moving storm, crossed the northern section of the Florida Reef Tract. Hurricane Andrew was a category 4 storm on the Saffir–Simpson hurricane damage-potential scale (AHRENS, 1993), making it the largest storm to hit South Florida in at least 50 years. Maximum sustained winds of 225 km h^{-1} with gusts of up to 285 km h^{-1} were measured 15 km north of Elkhorn reef (POWELL & HOUSTON, 1993; RAPPORT & SHEETS, 1993). It is estimated that winds were slightly less than maximum, about 210 km h^{-1}, directly over Elkhorn reef (Fig. 1).

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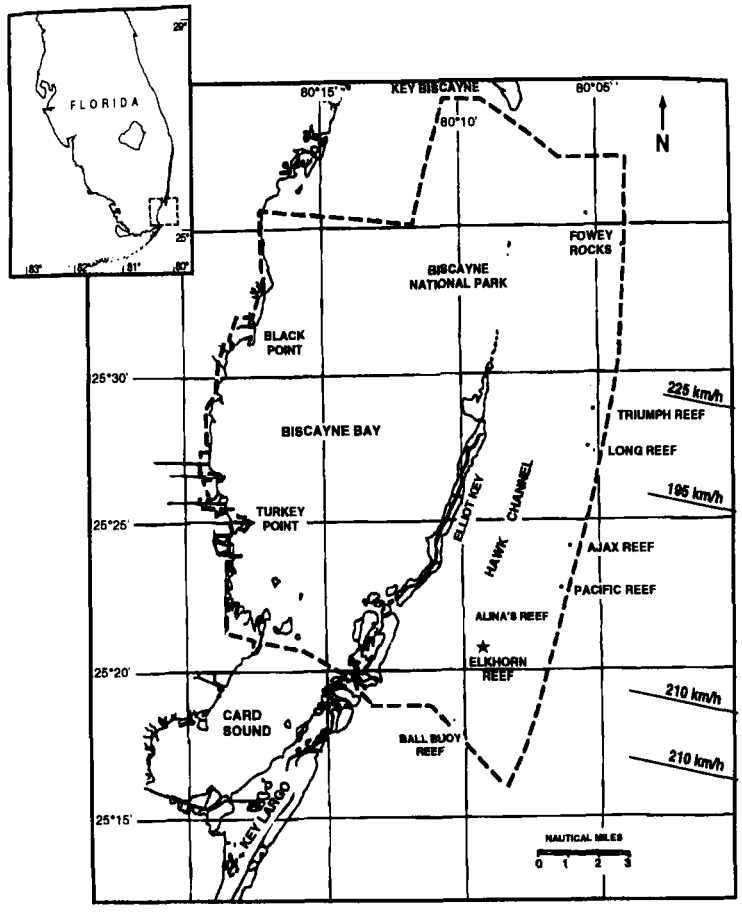

Fig. 1. Map of South Florida including the path of Hurricane Andrew with demarcations indicating bands of approximate maximum sustained winds, the boundaries of Biscayne National Park, and several of the larger reefs of the northern Florida Reef Tract. The inset shows the location of the study site, Elkhorn reef.

Damage to the reef from hurricane-generated waves was very patchy (DAVIS *et al.*, 1994; TILMANT *et al.*, 1994). The worst damage was inflicted to dense aggregates of *Acropora palmata* upon the shallow reef flat (LIRMAN & FONG, 1996a,b). The largest *A. palmata* patch on the reef was severely damaged, with >60% of the colonies broken into fragments and transported to the back-reef area. These fragments formed a field of live 'rubble' that originated on the shoreward boundary of the reef flat and extended roughly 15 m westward.

In this study, we defined 'fragments' as those broken colonies or branches that lacked a basal attachment to the substrate and were still covered by live tissue, regardless of size. Once a fragment reattached to the bottom, reducing the chance of being removed from the reef, it became a 'recruit'. Several months after Hurricane Andrew, we observed that many of these new recruits had developed small growth protuberances on their top and side surfaces. We termed these features 'proto-branches'.

2. Wound healing rates of damaged colonies

The most common type of damage observed after Hurricane Andrew was broken *Acropora palmata* branches. On 22 October 1992, an experiment was initiated to determine the rate at which *A. palmata*

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healed broken branch ends. We marked five large colonies, each with 5-8 injury sites, and took fixed-frame photographs of each lesion. On 14 December 1992 and 2 March 1993, injury sites on the surviving colonies were re-photographed. Sequential photographs of the lesions were projected onto a screen. Distances across several areas of each wound were measured, and using planimetry, the area of each wound estimated. From these sequential measures we calculated the healing rate as growth or expansion across the exposed skeleton as well as changes in wound area through time. The community that colonized the exposed skeleton was noted.

3. Distribution and survivorship of hurricane-generated *Acropora palmata* recruits

To quantify changes in the size-frequency distribution of fragments and recruits that were deposited in the rubble field, 5 parallel transects were selected that originated on the shoreward edge of the reef flat where the rubble field started, and bisected the largest *A. palmata* fragment field on the reef. Along each transect, we flipped a 1 m^2 quadrat from east to west until no more live fragments were found. The number of quadrats along each transect ranged from 8 to 12. Within each quadrat we counted the number of fragments, measured the size of each fragment as longest length and widest width, an noted whether the fragment was loose or cemented to the substrate. This survey was done in May 1993, 9 months after the storm, and repeated in March 1994, 19 months after the storm.

4. Distribution, abundance, and growth rates of proto-branches

In spring and early summer of 1993, many fragments had begun to regenerate. The process was initiated after most of the fragments had reattached to the benthos by accreting calcium carbonate skeleton between the fragment and substratum (Lirman & Fong, 1996a). By mid-June of 1993, small proto-branches appeared on the top and sides of nearly all of the surviving fragments.

On 1 July 1993 we marked 6 fragments, noted orientation of each fragment as a deviation from the upright pre-hurricane position, and measured the initial size both photographically and *in situ* as longest length by widest width. We mapped the distribution of proto-branches on each recruit and measured their maximum height. These measurements were repeated on 6 October 1993. Analysis of variance was used to determine if there were any significant differences between the average height of proto-branches among fragments for each date.

Two hypotheses were developed to investigate differences in distribution and growth of proto-branches among colonies. The first hypothesis was that there would be differences in number and height of proto-branches on recruits due to the degree of change in orientation that the fragment experienced as a result of transport during the hurricane. The underlying rationale is that more energy would have to be expended to initiate upward growth by a coral colony that was further from the original orientation, and growth would thus be slower on fragments with greater deviations. Marked and measured fragments were divided into 2 categories. One group was rotated 180 from their original orientation on the 'mother' colony; these fragments were standing on what were the branch tips of the original colony. The rest of the fragments were lying flat on the benthos, or shifted 90 from the original orientation. Differences in proto-branch height due to fragment orientation were tested with *t*-tests.

Our second hypothesis was that the initial size of new recruits would have a significant effect on the abundance and/or height of proto-branches: more energy should be available to allocate branch growth if more polyps were present on a recruit. We used regression analysis to quantify the relationship between initial fragment size (estimated roughly as area by multiplying length by width) and proto-branch number and height.

The next objective was to assess how spatial distribution patterns of new proto-branches would affect their height. The sketch of each coral fragment revealed distinct patterns of proto-branch distribution on each colony, with some proto-branches being aggregated, while others were isolated. Proto-branches were segregated into 'solitary' and 'aggregated' categories by defining an aggregate as containing proto-branches < 5 cm from each other, and a solitary proto-branch as > 5 cm from other proto-branch. The hypothesis is that clustering of newly generated proto-branches would reduce their average growth. *T*-

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tests compared the mean height of solitary vs. aggregated proto-branches and the height of solitary branches vs. the largest proto-branch within an aggregate.

Our last objective was to quantify the effect of being within an aggregate on the growth rate and resultant size of new proto-branches, *i.e.*, to quantify the patterns of variability within an aggregate. We hypothesized that, through time, one of the proto-branches within an aggregate would become dominant. This was based on two lines of reasoning. First, as most adult colonies have a central base, were reasoned that these aggregated and genetically identical proto-branches must eventually fuse under one dominant. The second argument is based on energy allocation. Under similar environmental conditions any two given 100 cm^2 patches of coral tissue supporting new proto-branches will be able to allocate roughly the same amount of energy to one solitary proto-branch as to an entire aggregate of 3 to 7 proto-branches. One proto-branch may thus develop faster and eventually over-top or shade the others. To investigate this variability we ranked proto-branches within 7 randomly selected aggregates from smallest to largest, and graphically compared average height among ranks for each sampling date using a one-way ANOVA followed by a Fisher's PLSD multiple comparison test. Lastly, the height of solitary proto-branches and the largest proto-branch within an aggregate were compared using a *t*-test.

Results

1. Wound healing rates of damaged colonies

Between the October and December sampling periods, the amount of disturbance to *A. palmata* increased considerably due to strong winter wind events (NWS, 1993; LIRMAN & FONG, 1996b). Three of the five colonies marked in October for our lesion study were completely shattered, with large pieces still bearing tags transported to new locations. For those *A. palmata* colonies that survived this secondary disturbance (two colonies with 6 identifiable lesions), the rate at which tissue regenerated and expanded linearly over the exposed skeleton slowed with time (Table 1). The healing rate was faster in the first period (October–December) than in the second (December–March). There was no relationship between the initial area injured and the mean linear distance the 'healing lip' of the coral tissue moved over each injury ($r^2 = 0.006$; $P > 0.05$); thus, the rate that the 'healing lip' advanced over a wound was not dependent on the initial size of the wound. Variability in the rate that the tissue moved over the injuries was great; both the highest and lowest rates were measured in the first time period. This variability occurred because wounds did not always heal evenly, with one edge of tissue moving further than others.

In October, green filamentous macroalgae colonized 90% of the injury sites. By December this algal community changed to turfs dominated by red filamentous algae. These algal turfs did not vary visually among injury sites. Except for lack of a crustose coralline component, the turf did not appear to differ from the ubiquitous turf community that occupies hard substrates on other parts of Elkhorn reef (LIRMAN, unpublished data). We did not observe colonization by sponges or coral recruits.

The area of tissue that covered the injury sites during the 5 month period (October–March) varied considerably between injuries (Table 1). There was a simple linear relationship between the initial size of the injury and the area of the wound that was covered by tissue during the interval ($r^2 = 0.87$; $P < 0.01$); larger wounds resulted in larger areas healed. There was a significant relationship between the perimeter-to-area ratio (P : A ratio) and both the area ($r^2 = 0.75$; $P < .05$) and

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Table 1. Wound-healing rates for six injury sites on two *Acropora palmata* colonies that remained standing 6 months after Hurricane Andrew. Lesions were marked and photographed with a fixed-distance framer on 22 October 1992 and 2 March 1993. Healing rates are reported as linear extension or planar distance that the tissue around the edge of a wound grew over the bared skeleton. Values are mean (± standard error) of n measurements of growth across the injury site (nd = no data). Areas were estimated by weighing paper cutouts of each injury photograph and transforming weight to area. Mean growth is in cm $month^{-1}$, areas are in cm^2, and area recovered in cm^2 $month^{-1}$.

colony no.	lesion no.	mean growth T1–T2	T2–T3	T1–T3	initial area [cm^2]	initial perimeter	perimeter/area	area recovered T1–T3	% area recovered T1–T3
3	2	1.80 (0.72) n = 3	1.32 (0.66) n = 3	1.91 (0.51) n = 7	48.1	48.8	1	2.3	21.1
3	3	3.89 (0.53) n = 2	1.24 (0.90) n = 3	2.42 (0.82) n = 4	105.3	56.2	0.5	6	24.8
3	4	2.38 (0.31) n = 10	0.83 (0.33) n = 10	1.46 (0.17) n = 10	12.6	19.9	1.6	1.2	42
3	5	0.28 (0.35) n = 10	0.19 (0.24) n = 10	0.41 (0.16) n = 10	69.6	65.8	0.9	3.6	22.3
3	6	nd	nd	1.97 (0.49) n = 7	88.7	51.2	0.6	3.4	17
4	4	nd	nd	1.50 (0.23) n = 8	11.8	19.3	1.6	1.6	59.3
overall mean (SE) =		1.59 (0.31)	0.78 (0.21)	1.47 (0.16)				3.0 (0.7)	31.1 (6.7)

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percentage ($r^2 = 0.75$; $P < 0.05$) of tissue regenerated. The P : A ratio provides an estimate of wound shape. Round wounds (low P : A ratio) healed a smaller percentage of the initial wound area than wounds with a flattened or compressed shape (high P : A ratio). Thus, the number of coral polyps along the edge of a wound that were available to bud and form new tissue was directly related to the amount of tissue regenerated.

2. Survival of hurricane-generated *Acropora palmata* fragments

In May 1993, 9 months after the storm, 218 fragments in 48·1 m^2 quadrats within the hurricane-generated field of fragments were measured (mean = 4.56 fragments · m^{-2}; S.E. = 0.54). Most fragments were longer than 5 cm (Fig. 2), possibly because few smaller fragments were generated or because fragments of this size had rapid mortality in the first 9 months. The greatest abundance of hurricane-generated fragments was in the next three 5-cm-wide size classes, with a longest length from 5 to 20 cm. Forty-nine percent of the fragments were 5–20 cm, 40% 20–50 cm, and 11% >50 cm. At this time, 72% of the sampled fragments had cemented to the substratum, becoming new recruits. There was no difference in mean size (longest length) of uncemented and cemented recruits (*t*-test; P = 0.56).

More fragments were recorded within our transects during the March 1994 survey than in the previous year; 271 fragments (mean = 5.24 fragments · m^{-2}; S.E. = 0.6). By this survey (19 months after the storm), 94% of the sampled fragments had become cemented recruits. Size did not significantly affect the rate of asexual recruitment: there was no difference in mean size of recruits and uncemented fragments (*t*-test; P = 0.80). Mean fragment size did not differ sig-

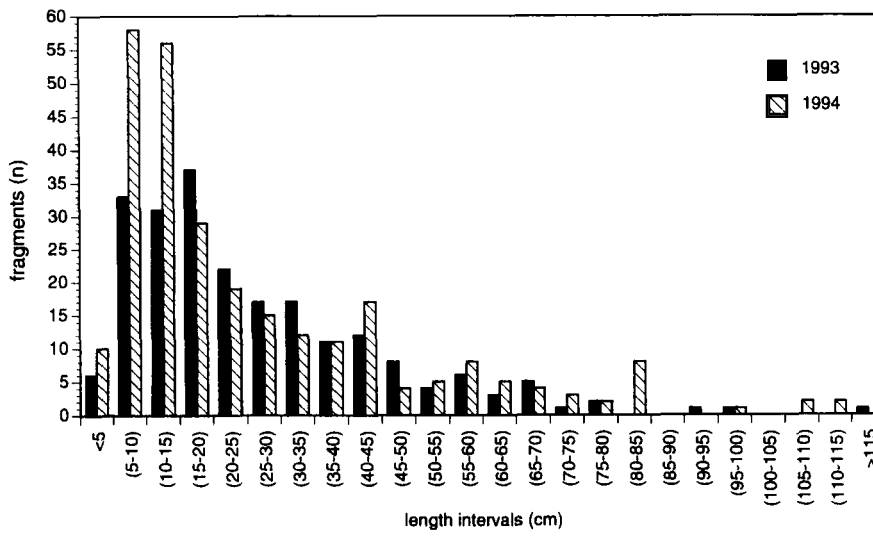

Fig. 2. Size-frequency distributions of coral fragments and recruits found in the 48 m^2 of hurricane-generated rubble surveyed in May 1993 and March 1994.

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nificantly between 1993 and 1994 (t-test; $P = 0.945$). The relative distribution of fragments in the small (5–20 cm), medium (20–50 cm), and large (> 50 cm) ranges were 53, 29, and 11, respectively (Fig. 2) in 1994. During our surveys we observed several small or medium-sized fragments that had developed continuous tissue connections among them, making them almost indistinguishable.

3. Distribution, abundance, and growth rates of proto-branches

The number of proto-branches on each of the coral recruits ranged from 17 to 36 (Fig. 3a). There were significant differences between the mean height of these branches on different coral colonies both in July (ANOVA; $P < 0.05$) and October (ANOVA; $P < 0.001$). However, the patterns were different in the two sampling periods. In July, the hierarchy for average proto-branch height was recruit $2 > 3 = 4 = 1 = 6 = 5$ (Fig. 3b, Fisher's PLSD). By October the hierarchy changed to $3 = 2 = 1 = 6 > 5 = 4$ (Fig. 3c, Fisher's PLSD). This change in hierarchy reflected different average growth rates (Fig. 3d, ANOVA $P < 0.001$). Growth rates of proto-branches on recruits 3, 1, and 6 were significantly faster than those on recruits 5 and 4, while the mean rate on recruit 2 was intermediate, differing significantly from only the fastest (3) and slowest (4) rates (Fisher's PLSD).

The average growth rate of proto-branches on colony 4 was negative. Polyps surrounding each proto-branch also grow, increasing the height of the tissue surrounding each proto-branch. Since net proto-branch growth is the difference between the growth of polyps on the branch and polyp growth on the surface of the recruit, a possible explanation for the observed negative growth is that the surrounding tissue grew faster than the proto-branch. GLADFELTER *et al.* (1979) also noted that the deposition of skeletal material on the arms of *A. palmata* sometimes exceeds that of the growing tips.

The final position of the recruit compared to its original orientation on the colony had a significant, but short-term, effect on proto-branch height (Fig. 4a). It significantly affected height in July (t-test, $P < 0.05$), but not in October (t-test, $P = 0.70$). In July, proto-branches on recruits that were lying flat (rotated 90°) were larger than those that were upside down (rotated 180°). By October, the difference was no longer significant. In contrast, the shift in orientation of the original branch did not affect the number of new proto-branches in July (t-test, $P = 0.42$) or October (t-test, $P = 0.54$). There were no differences between the mean proto-branch numbers on differently oriented corals (Fig. 4b), and their abundance did not change between sampling periods.

The size of a recruit affected the number of proto-branches on that recruit, with a significant linear relationship between initial area of a recruit and branch number ($r^2 = 0.855$; Fig. 5a). However, no significant relationships were found between initial size and proto-branch height ($r^2 < 0.05$; Fig. 5b).

Proto-branches that were distributed within an aggregate (< 5 cm apart) were significantly smaller than those that were solitary (> 5 cm from any other proto-branch; Fig. 6; t-test, $P < 0.05$ in July and October). In July, the mean height of the solitary category was 2.9 cm (S.E. = 0.2; n = 40) vs. 2.4 cm (S.E. = 0.1; n = 82) in the aggregated category. Although all branches grew between July and October,

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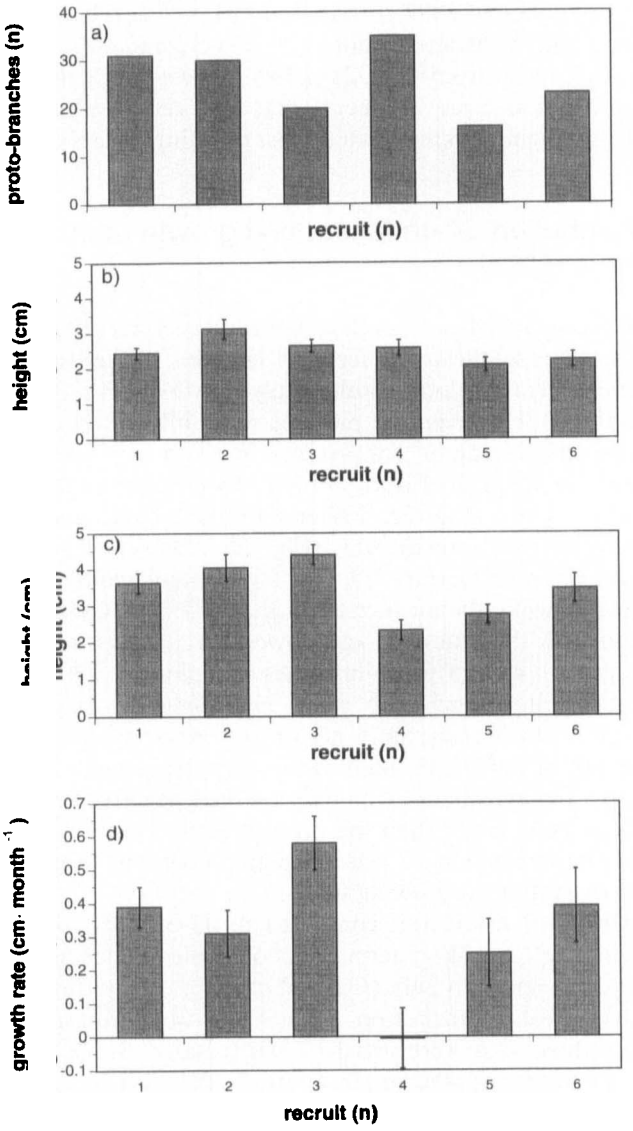

Fig. 3. Patterns of distribution and abundance of proto-branches among *Acropora palmata* recruits. Panels show: (a) number of proto-branches on each recruit; average proto-branch height on each recruit in (b) July and (c) October; (d) average growth rate of proto-branches between the two sampling periods. Bars are standard errors.

the mean height of the solitary types was still larger than that of aggregated types, 3.9 cm (S.E. = 0.3; n = 40) and 3.2 cm (S.E. = 0.2; n = 83), respectively. Solitary proto-branches grew significantly faster than those within an aggregation (t-test, $P < 0.05$): 0.53 cm · month^{-1} (S.E. = 0.05) versus 0.20 cm · month^{-1} (S.E. = 0.09). The variability of the growth rate within an aggregate was also much higher.

The effect of proximity to other proto-branches on the largest branch within an aggregate was determined by comparing heights of solitary and largest within-

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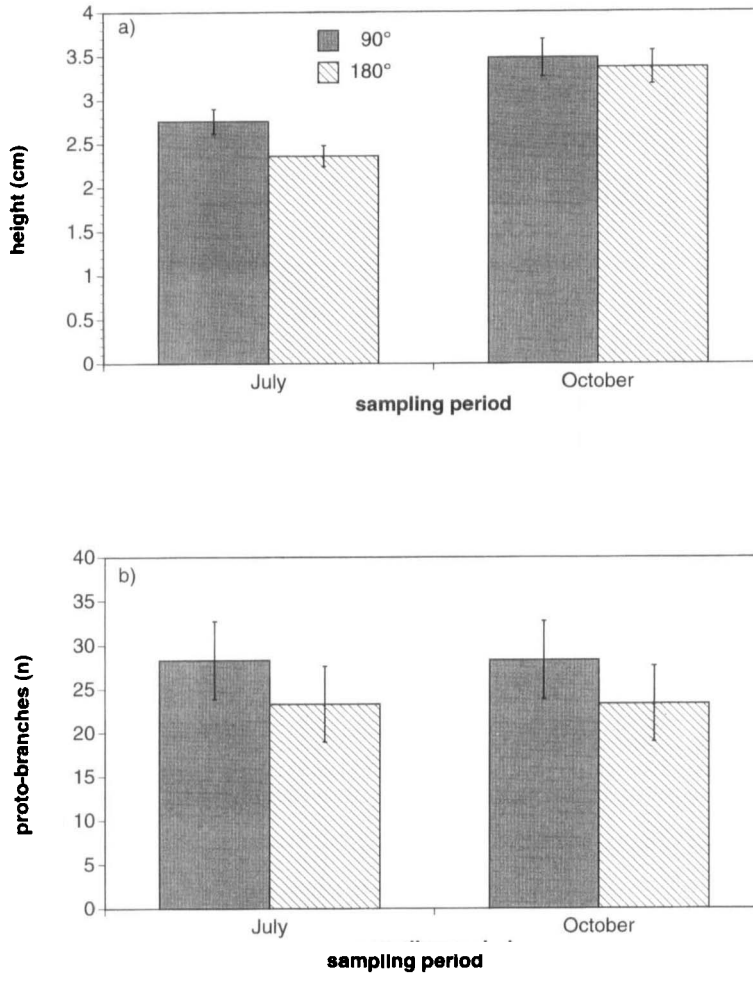

Fig. 4. Differences in (a) average height and, (b) number of proto-branches in relationship to the degree of rotation that the recruit experienced from the time it was attached to the 'mother' colony. Recruits were grouped into two categories. Recruits in one category were originally upright branches that ended up lying flat on the benthos (rotated 90). Recruits in the other category were sections of the mother colony that landed upside-down on the branch tips.

aggregate branches. The mean height of the latter was not significantly different from that of the solitary types in either July (t-test; $P = 0.60$) or October (t-test; $P = 0.56$). In July, the mean height of the largest within-aggregate branch was 2.9 cm (S.E. = 0.2), while solitary branches measured 3.1 (S.E. = 0.3). In October, the corresponding values 3.9 (S.E. = 0.3) and 4.3 (S.E. = 0.3), but this difference was not significant.

To test for the effect of being within an aggregate, such proto-branches were ranked from largest to smallest (Fig. 7a). There was a difference in the mean size among ranks in both July (ANOVA; $P < 0.001$) and October (ANOVA; $P < 0.0001$). In July, mean height within an aggregate ranged from 1.0 (S.E. = 0.0) in the smallest to 3.2 (S.E. = 0.2) in the largest rank. In October, the value in the

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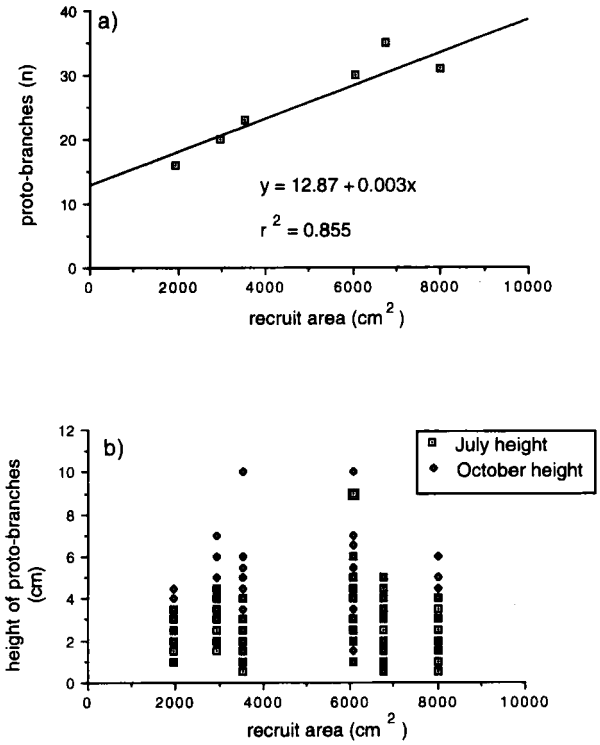

Fig. 5. Relationship between (a) proto-branch number and (b) proto-branch height and recruit size estimated as live tissue area (cm^2). Squares with inset diamonds in panel b signify overlapping data points.

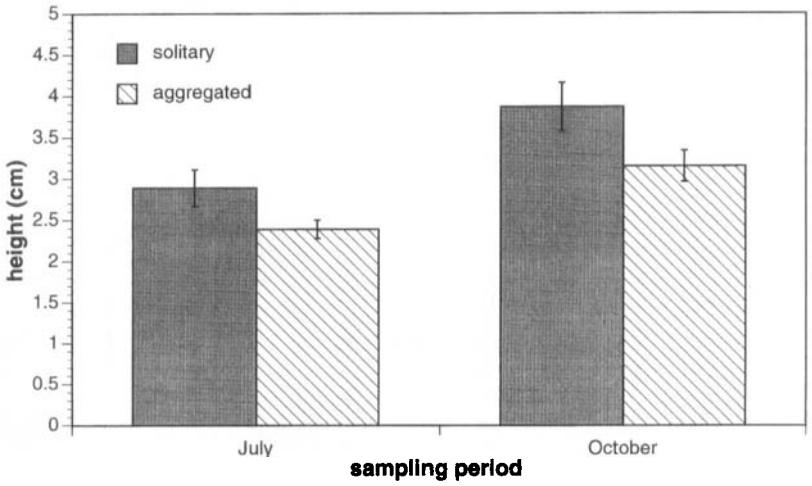

Fig. 6. Differences in average height of solitary and aggregated proto-branches in the two sampling periods. Solitary proto-branches were > 5 cm away from other proto-branches and aggregated proto-branches were > 5 cm apart. Bars are standard errors.

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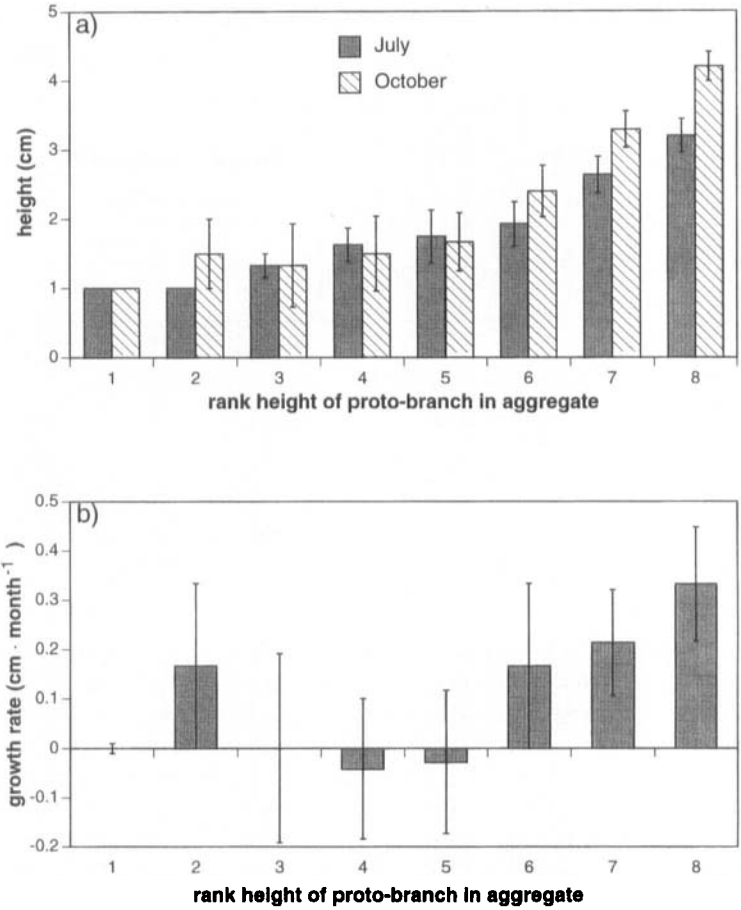

Fig. 7. Differences in (a) average height and (b) average growth rate of aggregated proto-branches in the two sampling periods. Proto-branches were ranked from smallest (rank 1) to largest (rank 8) based on height. Bars are standard errors.

largest rank was 4.2 cm (S.E. = 0.2), while it remained 1.0 cm (S.E. = 0.0) in the smallest rank. In both sampling periods, mean proto-branch height in the two largest ranks was significantly larger than in the other 6 ranks (Fisher's PLSD). Within the smallest 6 ranks, growth from July to October was extremely variable (Fig. 7b), ranging from −0.042 (S.E. = 0.142) to 0.167 (S.E. = 0.167). Mean growth rate within the largest two ranks was also variable, but higher than in the other ranks.

Discussion

Acropora palmata is adapted to disturbance, but has very different adaptive strategies for different disturbance regimes (Fig. 8). A chronic (high frequency), intermediate intensity disturbance regime (*e.g.*, reef flat zone) favours the maintenance of dense adult populations at relatively stable abundances. Adaptations to this

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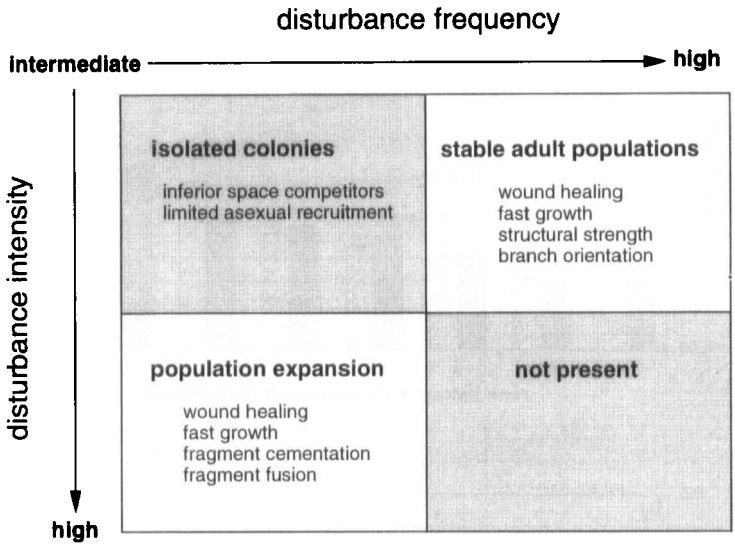

Fig. 8. Conceptual model of the adaptations of *Acropora palmata* to different disturbance regimes.

disturbance regime had been previously documented (SCHUHMACHER & PLEWKA, 1981; ROGERS *et al.*, 1982; BACK & CRIENS, 1981; BAK, 1983). When both frequency and intensity of disturbance are decreased (*e.g.*, deeper, more sheltered zones), the species can be found as isolated colonies that do not aggregate into dense patches. In sheltered areas, *A. palmata* colonies do not fragment as often, resulting in decreased asexual recruitment. This, coupled with *A. palmata*'s limited success in space competition (LANG, 1973; JACKSON, 1991), may explain the observed distribution. In this study, we documented that low frequency but high intensity disturbance (*e.g.*, hurricanes and tropical storms) favoured the rapid expansion of an *A. palmata* population by asexual recruitment of hurricane-generated fragments and persistence of injured adults by wound healing.

In contrast, *Acropora cervicornis*, another branching species believed to be adapted to high energy environments, has a different strategy for high intensity disturbance, sacrificing persistence of adults for new asexual recruits. This congener does not allocate as much energy as *A. palmata* to enhancing adult persistence, but may instead fragment completely forming many asexual propagules (GILMORE & HALL, 1976; TUNNICLIFFE, 1981; 1983; JACKSON & HUGHES, 1985). For example, prior to Hurricane Allen, 80% of the *A. cervicornis* colonies surveyed on transects across three zones of the reefs of Discovery Bay were formed from broken and recemented fragments (TUNNICLIFFE, 1981; 1983).

Wound healing is a trait that aids in coral recovery, and thus is an adaptation to any sort of a disturbance. BAK (1983) showed 100% regeneration of artificially produced small lesions (1 cm^2) within 80 days. Our study on naturally-produced lesions showed less recovery. One explanation may be that variability in size and orientation of these natural lesions affected growth. Alteratively, hurricane-damaged colonies may have healed each wound more slowly because of the extent of the colony injury. The average linear growth over the lesions produced by Hurricane Andrew is similar to the regeneration rates of *A. palmata* lesions marked

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after Hurricanes David and Frederic in St. Croix (ROGERS *et al.*, 1982). In our study, the shape of the resulting lesion (*i.e.*, the number of polyps on the periphery) was the most important factor affecting the regeneration process.

Our results supported other findings that healing rates of natural and artificially produced lesions on *A. palmata*, as well as other coral species, slowed with time (BAK & STEWARD VAN-ES, 1980; ROGERS *et al.*, 1982; BAK, 1983). Similarly, GLADFELTER (1991) showed that the areal growth of *A. palmata* 'crusts' decreased significantly during winter months. One possible explanation is that winter conditions (changes in temperature, light, and turbidity) reduced growth. However, it is not clear if these factors cause the reduced healing rate we recorded for injuries on Elkhorn reef. We measured the fastest growth in October through December, with slower growth from December through March, both periods when many of these factors are fluctuating.

A trait that promoted both recover and expansion of the *Acropora palmata* population following hurricane damage was the high rate of asexual recruitment from fragments. A large percentage of the hurricane-generated fragments reattached to the substratum, forming new recruits in areas that were not previously occupied by *A. palmata* and expanding the area covered or dominated by the population (LIRMAN & FONG, unpublished data). Although early mortality rates (0–9 months after the storm) were not measured, in the intermediate term (9 to 18 months) we observed that few fragments died completely. Fragments became smaller by partial mortality and larger by tissue expansion and fusion. The fusion of *A. palmata* fragments has been previously reported by BAK & CRIENS (1981) and ROGERS *et al.* (1982).

New recruits of *Acropora palmata* underwent rapid initial growth. On Elkhorn reef, the new growth was directed into upward growth rather than into rapid areal or planar expansion as was found after catastrophic predation on *Acropora* spp. by *Acanthaster planci* (COLGAN, 1987). This may be because *A. planci* predation leaves behind intact, upright skeletons, while hurricanes flatten them, and recovering old skeleton may cost less energetically than growing one. Planar growth or 'sheeting' was observed for sexual recruits of *A. palmata* growing on dead standing skeletons of the same species (JORDAN-DAHLGREN, 1992). Growth rates measured for the proto-branches (average = 3.6 $cm \cdot a^{-1}$) were slower than those recorded for branches on intact, undisturbed *A. palmata* colonies (4–11 $cm \cdot a^{-1}$; BAK, 1976; GLADFELTER *et al.*, 1978; ROGERS *et al.*, 1982; GLADFELTER, 1991). This slower growth may be partly due to the disturbance experienced by the fragments and the energy spent in stabilization and fusion.

Two years after the storm, it is still unclear what sequence of structural changes the recruits will undergo to achieve adult morphology; this sequence will have to be followed over a longer time scale. However, patterns of early growth were uncovered across several spatial scales, from among recruits to among proto-branches within an aggregate. The only factor that affected proto-branch abundance was the initial size (area) of the recruit, indicating there is a finite number of new branches that a given area of tissue can support. Interestingly, all of the new proto-branches initiated at about the same time in early summer, and no new ones formed in the 9 months between sampling periods. On the other hand, growth rate and subsequent height was controlled, at least in the short-term, by the orientation of the fragment. The longer-term growth rate of proto-branches was controlled,

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to some degree, by distribution within a recruit. This occurred in two ways. First, growth of solitary proto-branches was faster than of branches in aggregates, suggesting that the former may become primary stalks. In addition, the within-aggregate effect on proto-branch growth suggests that one or two branches within each aggregate may become dominant.

One interesting observation about the growth patterns of the new recruits was the similarity to patterns of growth in higher plants, especially trees. In a similar analogy, TUNNICLIFFE (1983) compared regeneration and growth of *Acropora cervicornis* to the 'weedy' habit or brambles. There are two main similarities between the growth of new recruits of *Acropora palmata* and trees. First, the proto-branches of *A. palmata* are all initiated at the same time, much like buds in a temperate tree, suggesting a seasonal or temporal cue. Alternatively, growth of recruits may be synchronized by the need for a recovery period before they can produce new growth. In addition, since a considerable amount of energy may be spent in the stabilization process, upward growth may be delayed until this is achieved. The second similarity to tree growth is that, within an aggregate, one of the proto-branches 'dominates', and the others have reduced growth rates. This is analogous to plants exhibiting apical dominance. In trees, this dominant-subdominant relationship is tightly regulated by growth hormones; the controlling factors in new *A. palmata* recruits are unknown. We hypothesize that, after sufficient growth, 'solitary' proto-branches may become 'aggregated' at a larger scale, and this pattern of one becoming dominant and subsuming the rest may be repeated until the Elkhorn colony has the typical adult morphology.

Several authors believe it unlikely that organisms can evolve adaptive responses to disturbances such as hurricanes, because they occur very seldom relative to generation times (PAINE, 1979; SOUSA, 1984). However, hurricanes may not occur infrequently for corals that have continuous growth and often reproduce by colony fragmentation. These traits may result in high generation overlap and the long-term persistence of a given genotype (POTTS, 1983). Thus, a given population may be exposed to several major disturbances during its lifespan. VOGL (1980) maintains that organisms and the communities in which they exist must be considered in terms of the species' lifespans and life cycles.

In addition, smaller-magnitude disturbances of the same kind as hurricanes (*e.g.*, seasonal storms, groundings) occur at regular intervals. In the case of *A. palmata* populations, the smaller-scale disturbances they encounter on a seasonal basis may have resulted in the evolution of features that may also aid recovery after major disturbance events such as infrequent hurricanes.

Summary

Initial recovery of a population of the branching coral *Acropora palmata* damaged during Hurricane Andrew (Biscayne National Park, northern Florida Reef Tract, USA) was documented. Recovery was measured in three ways: the rate of wound-healing of damaged adults, hurricane-generated asexual recruitment rate, and the pattern and rate of initial upward growth of new recruits. Immediately after the storm, wounds on surviving adult colonies — formed when branches were broken — healed rapidly (1.59 cm of linear growth per month); however, this

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growth slowed with time and few wounds healed completely within the 18 months of the study. In this study, a broken fragment was considered an asexual recruit when it stabilized by cementation to the substratum. Nine months after the storm, 72% of 218 hurricane-generated fragments of *A. palmata* had become cemented to the substratum. Within 18 months of the storm, the number of fragments in the same area had increased to 271, average fragment size was larger, and 95% had become recruits. This suggests that, although significant damage was occurring well after the hurricane, recovery via asexual recruitment was significant. Once cemented, recruits showed complex patterns of branch formation, including rapid growth of solitary proto-branches, differential growth of proto-branches dependent on recruit orientation, and dominance within aggregates of proto-branches. We present a conceptual model that suggest *A. palmata* is adapted to disturbances of both low intensity and high frequency (conditions typical of reef flat zones) and episodic high intensity and low frequency events (hurricanes and tropical storms).

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